COURSE SYLLABUS
MTH 253 – CALCULUS 3

1. GENERAL INFORMATION

Course name:	Calculus 3
Course name (in Vietnamese):	Vi tích phân 3
Course ID:	MTH 253
Knowledge block:	Elective - Math
Number of credits:	4
Credit hours for theory:	40
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	Calculus 1, Calculus 2
Prior-course:	
Instructors:	Nguyễn Đăng Khoa - Nguyễn Thị Hoài Thương
Academic year:	Semester 3, 2025-2026

2. COURSE DESCRIPTION

The course is designed to equip the students with the concepts, techniques, and applications of calculus of several variables, including partial differentiation, multiple integration, and vector calculus. The course provides mathematical background for courses in Computer Science. The course emphasizes all three aspects: conceptual understanding, computational and problem solving ability, and exposure to applications.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Recognize properties of geometry, functions and their graphs in three dimensional spaces	LO1, LO2

G2	Explain meanings of the concepts of limits, continuity, partial derivatives of functions of many variables. Use fundamental properties involving these concepts to resolve basic problems such as rate of changes, extrema, including problems in real-world settings.	LO1, LO2
G3	Explain the concept of multiple integrals. Use fundamental properties of integrals to resolve basic computational integral problems including repeated integrals and change of variables.	LO1, LO2
G4	Use the concepts of vector calculus in dimesions 2 and 3 to do basic calculations, including line integrals and Green formulas.	LO1, LO2
G5	Use differential and integral calculus to resolve application problems, and use computational softwares to assist	LO1, LO2

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Use basic tools of coordinate geometries, such as distance, norm, inner product, angle, orthogonal projections, cross product.	U
G2.1	Explain meanings of partial derivatives, directional derivatives, and describe them both verbally and using formulas	U
G2.1	State precisely key results and methods, including linear approximation, directional derivatives, extrema, and provide justifications for those results	T
G2.2, G5.1	Do direct calculations, including computing partial derivatives of functions given by formulas, linear approximations, directional derivatives, local extrema and classifications, absolute extrema in some cases	U
G2.3, G5.1	Combine understanding of topics and computational skills to solve basic problems, including finding rate of changes, finding tangent planes, directional of fastest increases, maxima of quantities	U
G3.1	Explain meanings of multiple integrals, line integrals, being able to describe them both verbally and using formulas. Sketch some ideas of surface integrals.	U

G3.2	State precisely key results and methods, including Fubini formula, change of variable formula for integrals, Fundamental theorem of line integrals, Green theorem, and provide justifications for those results.	U
G3.3	Do direct calculations, including double integrals on simple regions, triple integrals in some simple cases, change of variables to polar coordinates	T
G3.4, G5.2	Combine understanding of topics and computational skills to solve basic problems, including totals and averages of quantities on regions in the plane	T
G4.1	Do direct calculations, including line integral of vector fields on the plane, Newton-Leibniz and Green formula	T
G4.2, G5.2	Combine understanding of topics and computational skills to solve basic problems works of forces on a motion in the plane or in space	T
G4.3	Sketch ideas for surface integrals and analogies of Green formula to dimension 3	I

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Chapter 1: Differentiation 1.1 Spaces Geometry in R^3 Lines and planes	G1.1	Lecturing demonstration, discussion
	1.2 Functions of several variables Limits and continuity	G1.1, G2.1	Lecturing demonstration, discussion. Take-home group works on graphs.
2	1.3 Partial derivatives	G2.1, G2.2, G2.3	Lecturing demonstration, discussion.
	Tangent planes and linear	G2.1,	Lecturing

	approximations, higher partial derivatives	G2.2, G2.3	demonstration, discussion. Quiz
3	1.4 Vector functions The chain rule	G2.1, G2.2, G2.3	Lecturing demonstration, discussion
	Directional derivatives and the gradient vectors	G2.1, G2.2, G2.3	Lecturing demonstration, discussion. Quiz on directional derivative
4	1.5 Extremum of functions of several variables	G2.1, G2.2, G2.3	Lecturing demonstration, discussion.
	Lagrange multipliers	G2.1, G2.2, G2.3	Lecturing demonstration, discussion. Quiz on optimization
5	Chapter 2: Integration Integration over rectangles Integrals over general regions, Volume, Properties	G3.1, G3.2, G3.3	Lecturing demonstration, discussion
	Fubini formula	G3.1, G3.2, G3.3	Lecturing demonstration, discussion
6	Continued	G3.1, G3.2, G3.3	Lecturing demonstration, discussion
	Change of variables	G3.1, G3.2, G3.3	Lecturing demonstration, discussion. Quiz on Fubini formula.
7	Continued. Applications	G3.1, G3.2, G3.3	Lecturing demonstration, discussion

	Midterm Exam		
8	Chapter 3: Vector Calculus Line integrals	G4.1, G4.2, G4.3	Lecturing demonstration, discussion
	The Fundamental Theorem for Line Integrals	G4.1, G4.2, G4.3	Lecturing demonstration, discussion. Take-home group works on computation of integrals.
9	Green's Theorem	G4.1, G4.2, G4.3	Lecturing demonstration, discussion
	Surface integrals	G4.1, G4.2, G4.3	Lecturing demonstration, discussion. Quiz
10	Reviews	G5, G6	Lecturing demonstration, discussion

Practical Laboratory Session

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either in the lab or at home. The lab works consists of at least three grades.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	**Assignments**			**40%**
A11	Class participation, Quizzes	Small quizzes in class, questions for individual. Brief 20-minute quizzes will be given on any topics in any lecture covered and any reading material assigned up to the time the quiz is administered. Missed quizzes cannot be made up. About 2 or 3 in-class group works.	G1.1 to G5.2	20%
A12	Assignments, including in Problem Sessions	Homework and projects for individuals or groups. During problem sessions, students will be asked to present works on homework problems. Selected problems can be assigned to groups of students, to be presented in class. Works could be typed or scanned and shown using projectors. About 2 or 3 take-home group works.	G1.1 to G5.2	20%
A2	**Exams**	Each student will be responsible for completing a mid-term and a final examination. The final examination will be cumulative. No makeup examinations will be given.	G1.1 to G5.2	**60%**
A21	Midterm exam	Written exam	G1.1 to G2.3 and G5.1	20%

A22	Final exam	Written exam	G1.1 to G5.2	40%

7. RESOURCES

Textbooks

- Calculus, Concepts and Contexts, James Stewart, Thompson Brooks/Cole, 7th Edition, 2012

References

- Giáo trình phép tính vi tích phân, Bộ môn Giải tích, Khoa Toán - Tin học, https://sites.google.com/view/math-hcmus-edu-vn-giaitich
- http://www.calculus.org

Softwares

- Mathematical computation softwares: Maxima, or Maple, or Matlab, or Wolfram Alpha, or GeoGebra …
- Software for typesetting mathematics: LaTeX, or LyX.

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the faculty and the university.

Class Attendance and Participation

- Please be on time to class.
- Regular class attendance is strongly advised and is necessary for students to fully grasp many of the course concepts. Class attendance is checked indirectly through in-class quizzes, which may not be announced ahead.
- Students in attendance are expected to be active participants in the course. This participation includes: contributing to class discussions, providing insight into the class discussion topics, raising questions, and relating class material to personal experiences and other course topics.

- No makeup examinations or quizzes will be given. If you miss a quiz or an exam with legitimate reasons, either that exam will be excluded in determining your final grade, or the grade of your next exam will also be taken as the grade for the missing exam.
- All activities in the classroom are for studying the subject matter. In particular, using electronic devices for purposes other than studying is not allowed. Those not heeding this rule will be asked to leave the classroom immediately so as to not disrupt the learning environment.